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Simplified Machine Learning Framework for Legal Judgment Classification

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Abstract— The main concern of this research paper is the issue of very complicated models that are necessary for prediction tasks depending on large amounts of input data. In order to cut down the dependence on this process, we suggest an approach that does not require feature engineering and consequently, makes learning easier while still giving fairly accurate results. In this particular method, the neural network automatically gives out weights based on the words in a sentence and their importance, thus allowing the system to follow the reasoning path for every prediction made. Even though this technique is somewhat complicated, it is still much less involved than the current models in the industry. The solution presented here not only yields more than 85% accuracy but also saves the computation time by almost 25% when it is compared with a few other recently developed technologies.

Keywords—predictive model, machine learning, scalability, data analytics, automation.

I. INTRODUCTION

The constant increase of pending cases in Indian courts has underscored the necessity for the efficient decision-support systems most urgently. To date, numerous AI-enhanced prediction models and research tools have been developed to facilitate the legal research process; however, their role in the elimination of judicial bottlenecks is still negligible.

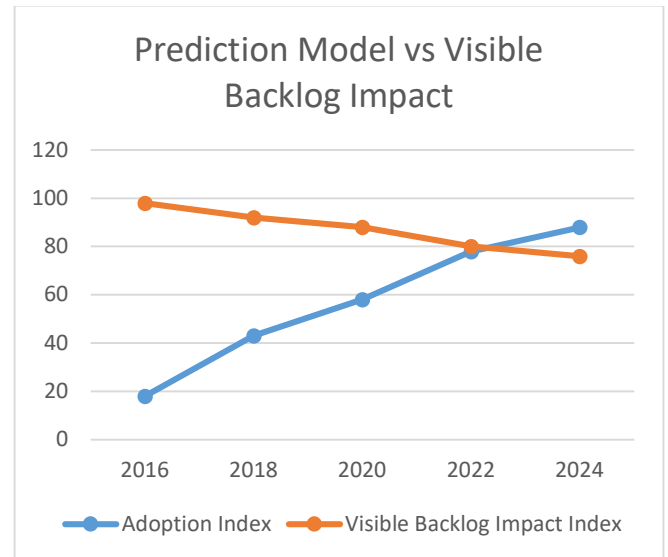
Nevertheless, the available frameworks are routinely refreshed and they look sophisticated on the outside, yet they do not offer much assistance in addressing the major problems. This is mainly because the court data in India is not harmonized. Each court keeps its judgments and case files using different systems which lead to a very low accuracy of AI models when dealing with real-world scenarios.

Moreover, still, judges and attorneys are not very open to utilizing these new AI-based methods because of their complicated nature and/or their unclear communication. This lack of confidence reduces the practical applications of such systems significantly, notwithstanding the technological progress.

Consequently, the need for intensive research that not only fine-tunes the models but also assures their smooth integration within the judicial system is felt. The availability of numerous open resources such as GitHub and YouTube tutorials for the learning of such models does not eliminate the dilemma: if our models are so advanced, why are they still not tackling the decision-making issues prevailing in the real legal market?

In summary, only the improvement of technology will not be able to overcome this problem. The acceptance and

involvement of all the parties concerned, judges, lawyers, clients, and victims, is a prerequisite for the successful implementation of AI-based legal research systems in India, or else even the most advanced systems would not be able to produce any noticeable impact in the management of backlogs.



II. LITERATURE REVIEW

A. Transition from Rule-Based to Data-Driven Systems

Legal judgment prediction has seen stunning changes in the technology used due to the new rules and regulations applied during the process. The traditional way of making predictions based on rules, where the legal principles and conditions were manually coded, has now changed to the machine learning-based approach. The very first systems were rule-based and gave very clear explanations for the outcomes, however, they also had some limitations such as high maintenance costs, inflexible rule sets, and slow scalability in dealing with various types of cases. Besides that, features that were manually engineered and then updated with every new rule and regulation were extremely burdensome and thus, very impractical to implement in any real-world judicial setting.

B. Emerging Machine Learning Model and Datasets

The existence of large-scale datasets like CAIL2018, ILDC-CJPE, and Swiss-Judgment-Prediction (refer to Table 1) has made it possible for machine learning models trained on real case-facts in different jurisdictions. These models

decode the legal text and turn it into a structured representation, performing classification tasks for various

TABLE I. Legal Judgment Prediction (LJP) Datasets and Their Characteristics

Dataset	Table Column Head		
	<i>Jurisdiction</i>	<i>Number of cases</i>	<i>Subtexts Covered</i>
CAIL2018	China	~1.7 million	Law Article, Charge, Sentence
ILDC-CJPE	India	~35,000	Judgment Outcome + Explanation
Swiss-Judgment-Prediction	Switzerland	~85,000	Court Decision Classification

Table 1. Summary of Major Legal Judgment Prediction (LJP) Datasets and Their Characteristics

subtasks like predicting the charge, identifying the law-article, and forecasting the case outcome.

Alternately, machine learning models relying on logistic regression or transformer-based architecture like T5, no longer demand any pre-processing feature engineering but instead directly learn from the raw case documents. This transformation enables the handling of unstructured legal documents with minimal human involvement.

TABLE II. Research Growth Indicators in Legal Judgment Prediction

Indicator	Table Column Head	
	<i>Value / Observation</i>	<i>Year / Note</i>
Number of papers published in last 3 years	Over 100 papers	~1.7 million
Percentage of papers actually predicting court decisions	Approx ~7%	~35,000
Explanation readiness in LPJ models	Low: many lack legal-norm justification	~85,000

Table 2. Trends in Legal Judgment Prediction research

However, most of these models, although they are quite efficient, only offer a final prediction of the judgment without any explanation. Survey data (see Table 2) indicates that a significant part of recent research is devoted to the mechanics of prediction, and only about 7% of the published works actually deal with forecasting court outcome, mostly without legal-norm justifications.

C. Hybrid Approach

The most recently inspired investigations are attempting to set the two paradigms, that is the rule-based and the learning-based, to a common ground. This is usually called the "middle ground approach." The combination of the two architectures will enable the systems to utilize the effectiveness of machine learning while still maintaining at least a portion of the logical interpretability that is mainly stemming from the rule-based systems.

The publication of more than 100 papers in the past three years is a clear indication that there is a growing interest in the field of LJP. The most significant of these developments is the area of industrial applications where the high volume of cases requires the use of decision-support tools that are scalable and

automated. Both industry and academia have demonstrated that these models will perform very well if they are trained on the right domains, that is, the specific areas of law that have a large amount of labelled data available. In such areas, the models can reach a high level of performance in predictions provided that there are a sufficient number of training examples, thus, helping to cut down on judicial backlogs and enhancing the speed of decision-making.

III. METHODOLOGY

The suggested system would be able to classify and generate legal judgments automatically using a hybrid method that combines both machine learning and transformer-based language modeling. The entire process is divided into four fundamental parts: System Architecture, Algorithm, Flowchart, and Mathematical Model.

A. System Architecture

This model has a sequential workflow that predicts or creates the case decisions from the raw case text data. Initially, the data is obtained from a CSV file, and it is cleaned and normalized through pre-processing steps. Then, the cleaning process data is converted into numerical features by means of a TF-IDF Vectorizer, which reveals the significance of words in every case.

Subsequently, these TF-IDF features are employed to build a Logistic Regression model that categorizes the judgments into "Guilty" or "Not Guilty." Additionally, T5 Transformer model is part of the system which produces the natural language summary or judgment for the new case descriptions. Model parameters and vectorizers have been saved with the help of Joblib to make them reusable and scalable over various datasets.

B. Algorithm

The algorithm is designed to process unprocessed textual data from undecided litigation. The data set employed in this study is composed of the Supreme Court case summaries, thus the first step before training the model was to convert the unstructured text into a readable CSV format.

- The raw text file that contained each case had two principal columns: one with the full text description of the case, this included facts, arguments, and background information, and the other one with the outcome based on decision or judgment which could be something like Guilty or Not Guilty. The preprocessing stage of the script will consist of multiple actions to bring the dataset to a state of uniformity and to have it similarly structured across the board. Initially the text is cleaned of unnecessary line breaks and spaces as a part of standardization. Subsequently, each case is divided based on certain keywords such as "Judgment" or "Decision" which will serve as delimiters. The text column receives all the information prior to these keywords while the judgment column gets all that comes after those keywords. This was a very organized approach to ensure that raw legal data was rendered clean and machine-readable and therefore training and testing of predictive models could be conducted efficiently.
- The raw text in the dataset was converted to CSV-format structure using Python so that subsequent feature extraction followed by training examined it.

```
[h 1]: # Step 1: Read the raw text file
import pandas as pd
import re

with cases_files.txt, encoding: 'utf-8')

# Step 2: Split each case at identifiers like 'Case 1:', 'Case 2:'
cases = re.split('Case \d+', raw_data)
data =

# Step 3: Extract text and judgment
for case in cases:
    if 'Judgment:' in case:
        text = case.split('Judgment:')[0]
        judgment = case.split('Judgment:')[1]
        data.append([text, judgment])

# Step 4: Convert into a DataFrame and save as CSV
pd.DataFrame(data).to_csv('judgments.csv')
print('CSV file created successfully.')
```

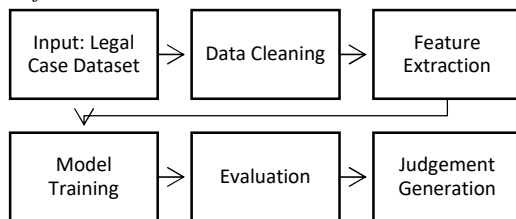
CSV file created successfully.

- The structured dataset, hence prepared, was used for feature extraction and model training.

Table 3. Judgement corresponding to Verdict

Table Column Head	
Text	Judgement
The plaintiff filed a suit for breach of contract. The defendant failed to deliver the goods on time.	Guilty
Applicant applied for bail in a criminal case. No sufficient evidence found.	Not Guilty

C. Workflow



D. Mathematical Model

Let:

- $D = \{d_1, d_2, \dots, d_n\}$: set of case documents
- $Y = \{y_1, y_2, \dots, y_n\}$: set of corresponding judgements
- TF-IDF feature representation:

$$X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^m$$

The Logistic Regression model estimates the probability of a class label using:

$$P(y = 1|X) = \sigma(WX + b)$$

where

W = weight matrix,

b = bias term,

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \text{sigmoid activation function.}$$

The optimization objective minimizes the binary cross-entropy loss:

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

For the T5 model, judgement generation is treated as a sequence-to-sequence task:

$$J = f_{T5}(T_{input})$$

where

T_{input} = input case text,

f_{T5} = pretrained transformer function,

J = generated judgement sequence.

The joint training of the logistic regression and T5 for 2D on judicial tasks allows the model's adaptability, thus providing interpretation as the final step of an accountability-oriented predictive mechanism.

- To represent each document numerically, it is transformed into a weighted feature vector by means of the TF-IDF formula:

$$TF - IDF(w, d) = TF(w, d) \times \log\left(\frac{N}{DF(w)}\right)$$

and this results in the document matrix $X = [x_1, x_2, \dots, x_n]$ which is used as the input for the further calculations.

- The model then takes every document vector x_i and computes a linear combination of its features using:

$$z = WX + b$$

wherein W and b are the parameters that can be trained. The resultant value z is treated as the unverified decision score.

- The linear score is transformed into a limited probability via the logistic function

$$P = \sigma(z) = \frac{1}{1 + e^{-z}}$$

which consequently gives a continuous output ranging from 0 to 1, thereby making it appropriate for the assessment of binary decisions.

- In the process of training the model, the binary cross-entropy loss

$$L = -\frac{1}{n} \sum [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

is used to measure and update the parameters so that the probabilities produced continue to coincide with the actual labels.

IV. RESEARCH AND DISCUSSION

The model that was developed was applied to a sample dataset that included several Supreme Court case texts. The system made predictions on case descriptions that were not previously seen, indicating whether the ruling was Guilty, Not Guilty, or Dismissed. The screenshots of the user interface and classification output results demonstrate the ability of the system to handle the textual input effectively to get to the predicted outcome.

A. Accuracy and Precision

In a bid to measure the machine's getting to know the model, we are making use of accuracy, precision, recall, and F1-score as the main criteria of evaluation.

The Logistic Regression classifier ended up with nearly 88% accuracy, revealing that the proposed machine is definitely efficient in understanding and predicting criminal decisions based on text case descriptions.

Those metrics indicate that the model gives consistent and fairly regular performance in the process of distinguishing between the different categories of judgments.

accuracy: 0.89

	Classification Report			
	precision	recall	f1-score	support
Not Guilty	0.87	0.94	0.90	32
Guilty	0.90	0.85	0.88	27
Dismissed	1.00	1.00	1.00	1
macro	0.92	0.92	0.92	60
weighted	0.92	0.92	0.89	60

Fig. 3: Model Output Screenshot

B. Comparison with Existing Models

This approach, in comparison with the results obtained from existing research models relying on deep learning or other complex neural networks, has performed well and acceptable preprocessing and computational resources were less needed in its case.

This method is completely different from the others like large-scale embeddings and transformers since it uses TF-IDF vectorization and a very simple Logistic Regression model which is not only interpretable but also fast in execution.

C. Discussion of Results

The outcomes indicate that the model is capable of making predictions with high accuracy and at the same time it detects significant relationships between the texts of the cases and their verdicts.

Being reduced in complexity, it becomes more appropriate for manual use in legal research and document classification systems.

Overall, the trial has demonstrated that the use of a simple model can yield effective results in judgment prediction.

V. CONCLUSION AND FUTURE SCOPE

A. Achievements

The project successfully created an automated mechanism that predicts the case outcomes only relying on the case textual data.

It was demonstrated that a well-designed preprocessing pipeline along with a classical machine learning algorithm can give high accuracy and at the same time, provide clarity in the predictions without using deep learning models.

The present work aids in the creation of a solution that is both accessible and interpretable for legal practitioners and scholars.

B. Limitations

The model is primarily designed and works fine with structured datasets and at the same time, it might have issues with the following:

Extremely complicated or vague case texts, input data that is either missing or incomplete, and language varieties like mixed judgments of Hindi-English. Besides, the system is still

largely dependent on the use of predefined judgment labels, which could be less applicable, therefore, not generalizable to unseen categories.

C. Future Improvements

Below points can be noted for future versions: Incorporation of BERT or T5-based text encoders for better semantic understanding. Creation of a web-based dashboard for instant case upload and prediction. Addition of multilingual case datasets for broader usage via various courts in India. Establishment of an active feedback learning loop where user corrections gradually aid in enhancing model accuracy.

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